# RRC-Raptor-V4H

- SDK Toolchain

V0.0.1

# Contents

# i. VERSION LIST

| Description | Version | Date | Author |
|---|---|---|---|
| SDK Toolchain | V0.0.1 | 2024/08/30 | kevinlin |
| | | | |

# ii. HASHTAG

#V4H
#BSP
#YOCTO
# Toolchain

# 1. Overview

Target :

Mainly for the RRC-Raptor project using Kernel 5.10.147 environment under the YOCTO 3.1.11 version to build the SDK and compile the environment.

# 2. System Requirements

PC software:

| | |
|---|---|
| Distributor : | Ubuntu |
| Description : | Ubuntu 20.04.4 LTS |
| Release : | 20.04 |
| Codename : | focal |
| Kernel : | 5.15.0-43-generic |

PC hardware:

| | |
|---|---|
| PC : | Intel i9-11900K @ 2.50GHz |
| Width : | 64 bits |
| Memory : | 64G |
| Block device : | 7.3TB Disk |

# 3. SDK Toolchain build setup

Information :

Use YOCTO i.MX Yocto-5.15.71-2.0.0 version。

Download and Build environment :

```
$ mkdir imx-yocto-bsp
$ cd imx-yocto-bsp
$ repo init -u https://github.com/nxp-imx/imx-manifest -b
imx-linux-kirkstone -m imx-5.15.71-2.0.0.xml
$ repo sync



$ DISTRO=fsl-imx-xwayland MACHINE=imx6dlsabresd source
imx-setup-release.sh -b build
```

## 3.1. SDK Toolchain Building

Step :

In the YOCTO compilation directory.

Specify toolchain recipes to execute compilation instructions.

```
$ bitbake meta-toolchain
```

After completion, the file will be built in <build_directory>/tmp/deploy/sdk/
poky-glibc-x86_64-meta-toolchain-core2-64-qemux86-64-toolchain-3.1.11.sh

toolchain Version :

glibc 3.1.11

gcc core2-64

### 3.1.1. SDK Toolchain Information

Execution information :

```
$ ./poky-glibc-x86_64-meta-toolchain-core2-64-qemux86-64-toolchain-3.1.11.sh -h


        Usage: {FILE}.sh [-y] [-d <dir>]
        -y              Automatic yes to all prompts
        -d <dir>        Install the SDK to <dir>
        ======== Extensible SDK only options ============
        -n              Do not prepare the build system
        -p              Publish mode (implies -n)
        ======== Advanced DEBUGGING ONLY OPTIONS ========
        -S              Save relocation scripts
        -R              Do not relocate executables
        -D              use set -x to see what is going on
        -l              list files that will be extracted
```

## 3.2. SDK Toolchain

Install :

Go to the working environment platform to be executed and install the toolchain under /opt/poky/3.1.11. You can use the following commands to set some environment variables.

```
$ ./poky-glibc-x86_64-meta-toolchain-core2-64-qemux86-64-toolchain-3.1.11.sh
-d /opt/poky/3.1.11
```

The toolchain required for the entire compilation will be installed.
After completion, you can see that there are some files placed under /opt/poky/3.1.11, some are environment variable configuration files, and the toolchain is all in the sysroot folder.

## 3.3. Compile

example :

If the SDK is installed in /opt/poky/3.1.11

We need to setup environment in a particular shell. You have to do it once for every new shell window you open :

```
$ . /opt/poky/3.1.11/environment-setup-core2-64-poky-linux
```

Compile hello_world.c :

```
$ $CC hello_world.c -o hello_world
```

Check executable file info :

```
$ file hello_world
```

# 4. APPENDIX